Sandith Ganhewage

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Education

Duke University — Durham, NC

Bachelor's in Computer Science and Electrical & Computer Engineering — Expected Grad: May 2027

- GPA: 4.00/4.00, Dean's List with Distinction (F24, S25), J. Welch Hariss Scholar
- Relevant Coursework: Engineering Design & Communication, Data Structures & Algorithms, Computer Architecture, Linear Algebra, Fundamentals of Electrical & Computer Engineering, Elements of Machine Learning

Experience

Founding Software Engineer — Sep. 2025 – Present

Finvase — Chapel Hill, NC

- Built AI-powered system for investment banking, enabling context-aware workflows to automate repetitive tasks
- Integrated Dropbox via OAuth for seamless data room migration, vectorizing files to enhance AI tool performance

Software Engineering Intern — Aug. 2025 – Sep. 2025

narb Inc. — Remote

- Designed user application for multi-modal LLM technology, utilizing Convex for a 30% increase in development speed
- Developed in React and React Native to develop a cross-platform front end solution, increasing overall usability
- Migrated existing applications to a new cross-platform framework prioritizing rapid development and implementation

Embedded Systems Engineering Intern — May 2025 – Aug. 2025

Qorvo Inc. — Greensboro, NC

- Designed an embedded systems solution to chip sorting with the Synax S9 Handler, reducing configuration time by 25%
- Deployed company-wide solution for 50+ employees through an intranet-hosted web interface with React and Node.js
- Implemented low-level Arduino GPIB interface, alongside a Raspberry Pi connecting to universal web UI via bluetooth

Undergraduate Researcher — Jan. 2025 – May 2025

Duke Center for Computational and Evolutionary Intelligence — Durham, NC

- Tested and researched machine learning model quantization and distillation techniques to reduce model size by up to 30% and inference latency by up to 10% while maintaining model accuracy using PyTorch libraries
- Researched applications in 5 established bioinformatics algorithms for hardware acceleration by tri-state CAMs

Projects

MeetUp | *React, React Native, Tailwind CSS, Convex, Clerk, Google/Outlook OAuth*

- Built cross-platform scheduling app with Google/Outlook calendar integration, simplifying event coordination
- Implemented dynamic group scheduling and unified profiles to accelerate onboarding and collaboration
- Secured user authentication with Clerk and designed privacy controls for configurable data sharing across groups

SmartStudy | *React, Node.js, MongoDB, Vercel, Tailwind CSS, Axios, FastAPI*

- Developed full-stack study platform enabling users to upload materials and generate 5+ types of AI-powered content
- Built FastAPI service with open-source LLM, reducing model size by 20% via chunking and prompt engineering
- Integrated MongoDB for secure storage and authentication, and deployed on Vercel with protected server-side features

Home Lab | *Debian, SSH, Tailscale, Docker, CasaOS*

- Designed fully functional home server for 20% of the cost of commercial products utilizing headless Debian Linux system
- Configured remote server access with Tailscale VPN, implementing exit node for secure browsing on public networks
- Implemented user-friendly web UI, allowing for account-based access to web applications hosted by docker containers

Extracurriculars

Duke Law School – *Instructional Technologies Consultant*, Durham, NC — Aug. 2025 - Present

narbhacks – *Hackathon*, Remote — Jul. 2025

Duke Catalyst – *Pre-professional Tech Society*, Durham, NC — Jan. 2025 – Present

Duke Applied Machine Learning – *Hardware Division*, Durham, NC — Sep. 2024 – Dec. 2024

Joint School of Nanoscience and Nanoengineering – *Research Intern*, Greensboro, NC — Jul. 2023 – Jan. 2024

Technical Skills

Languages: Java, Python, Javascript/Typescript, C, C++, HTML/CSS, MIPS

Technologies: React/Next.js, shadcn, Tailwind, Node.js, Docker, MongoDB, PostgreSQL, Firebase, Vercel, AWS

Skills: Data Structures & Algorithms, Git/Github, Machine Learning, Embedded Systems, CI/CD, System Design